



Topic 18

Direct-Current Circuits

Reading Question 18-1 (1 of 2)

In order to calculate the fraction of the total power delivered to the load resistor in a circuit consisting of a battery and a load resistor, one needs to know only the

1. load resistance.
2. internal resistance.
3. load resistance and internal resistance.
4. emf, load resistance, and internal resistance.

Reading Question 18-1 (2 of 2)

In order to calculate the fraction of the total power delivered to the load resistor in a circuit consisting of a battery and a load resistor, one needs to know only the

1. load resistance.
2. internal resistance.
- 3. Answer: load resistance and internal resistance.**
4. emf, load resistance, and internal resistance.

Reading Question 18-2 (1 of 2)

A battery placed in circuits with different load resistances can be considered as a source of constant

1. emf.
2. current.
3. terminal voltage.
4. nothing.

Reading Question 18-2 (2 of 2)

A battery placed in circuits with different load resistances can be considered as a source of constant

1. **Answer: emf.**
2. current.
3. terminal voltage.
4. nothing.

Reading Question 18-3 (1 of 2)

As more resistors are added in series, the equivalent resistance of the circuit approaches infinity. In contrast, as more resistors are added in parallel, the equivalent resistance

1. approaches infinity.
2. approaches zero.
3. becomes zero.
4. approaches $1\ \Omega$.

Reading Question 18-3 (2 of 2)

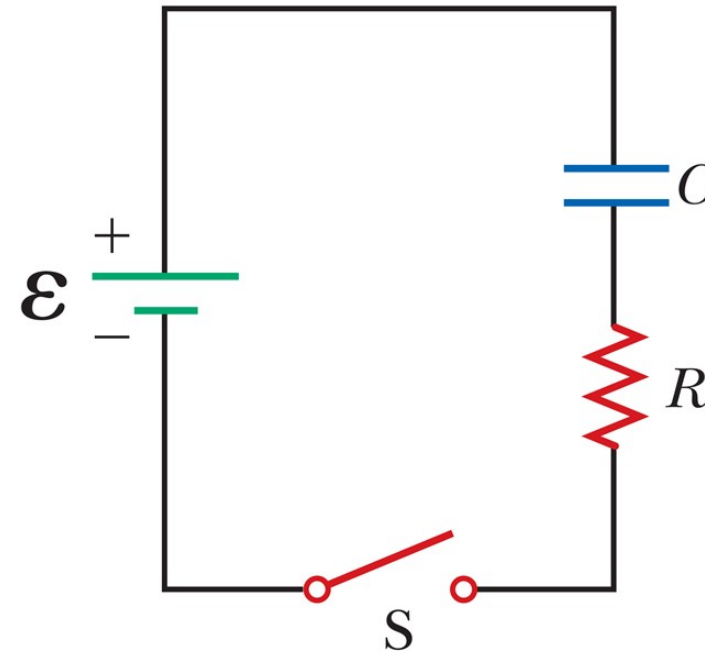
As more resistors are added in series, the equivalent resistance of the circuit approaches infinity. In contrast, as more resistors are added in parallel, the equivalent resistance

1. approaches infinity.
- 2. Answer: approaches zero.**
3. becomes zero.
4. approaches $1\ \Omega$.

Reading Question 18-4 (1 of 2)

At the moment the switch closes on this freshly connected RC circuit:

1. $Q = 0, I_0 = \varepsilon/R$
2. $Q = C\varepsilon, I_0 = 0$
3. $Q = C\varepsilon, I_0 = \varepsilon/R$
4. $Q = 0, I_0 = 0$



Reading Question 18-4 (2 of 2)

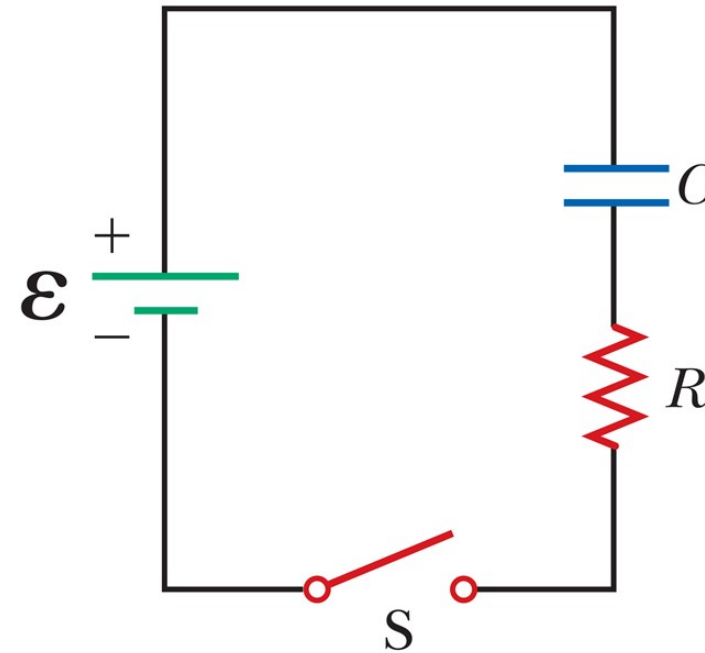
At the moment the switch closes on this freshly connected RC circuit:

1. Answer: $Q = 0, I_0 = \varepsilon/R$

2. $Q = C\varepsilon, I_0 = 0$

3. $Q = C\varepsilon, I_0 = \varepsilon/R$

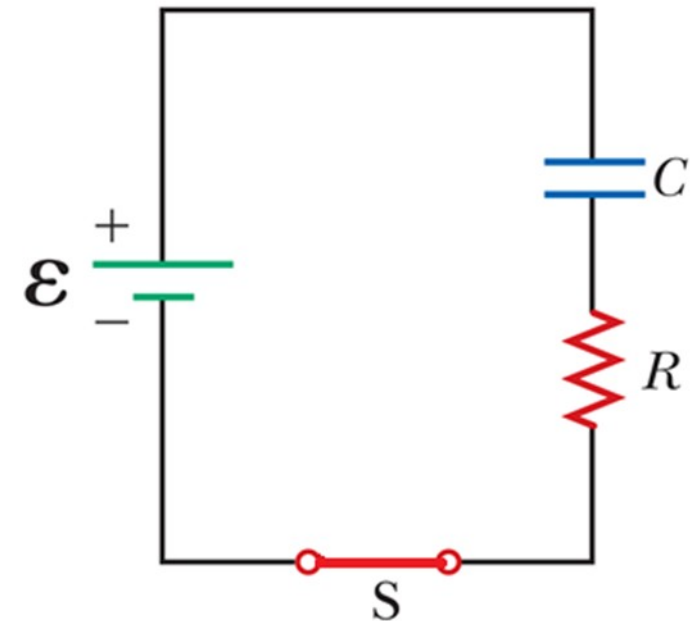
4. $Q = 0, I_0 = 0$



Reading Question 18-5 (1 of 2)

As the capacitor in the RC circuit reaches its maximum charge,

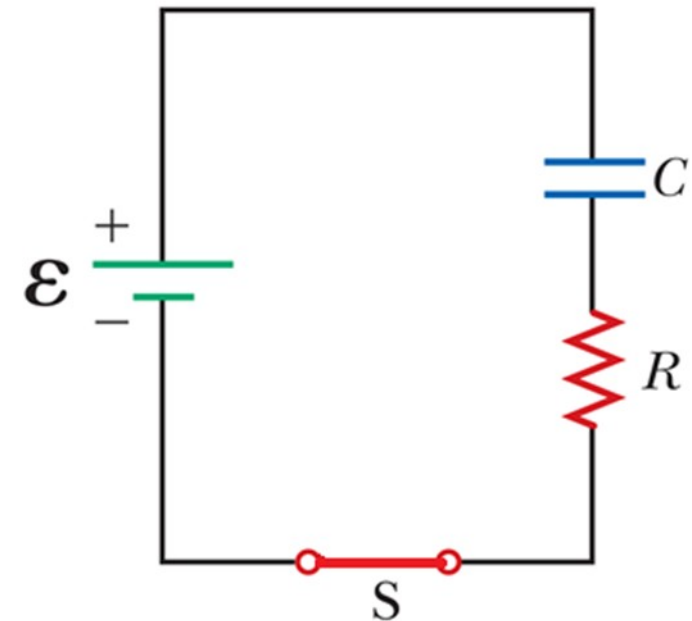
1. the rate at which the current changes decreases and the rate at which the charge changes increases.
2. the rate at which the current changes increases and the rate at which the charge changes increases.
3. both rates decrease.
4. both rates increase.



Reading Question 18-5 (2 of 2)

As the capacitor in the RC circuit reaches its maximum charge,

1. the rate at which the current changes decreases and the rate at which the charge changes increases.
2. the rate at which the current changes increases and the rate at which the charge changes increases.
3. **Answer: both rates decrease.**
4. both rates increase.



Reading Question 18-6 (1 of 2)

Kirchhoff's loop rule is equivalent to which of the following principles?
approaches infinity.

1. conservation of charge
2. conservation of energy
3. conservation of mass
4. conservation of force

Reading Question 18-6 (2 of 2)

Kirchhoff's loop rule is equivalent to which of the following principles? approaches infinity.

1. conservation of charge
- 2. Answer: conservation of energy**
3. conservation of mass
4. conservation of force

Reading Question 18-7 (1 of 2)

Appliances with three-pronged plugs compared to those with two-prong plugs are conservation of charge

1. safer to use and connected to ground.
2. no safer to use and ungrounded.
3. more dangerous to use and connected to ground.
4. safer to use and ungrounded.

Reading Question 18-7 (2 of 2)

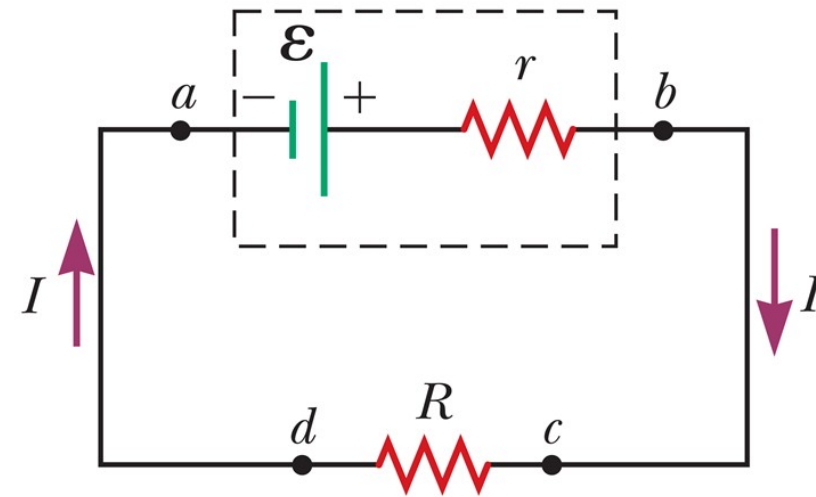
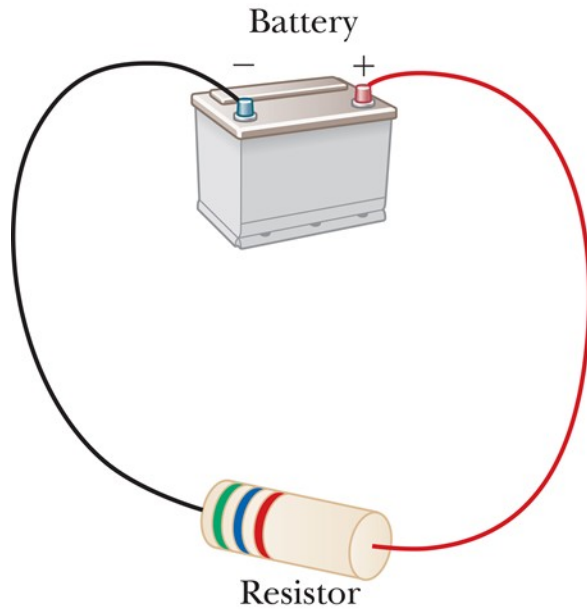
Appliances with three-pronged plugs compared to those with two-prong plugs are conservation of charge

1. **Answer: safer to use and connected to ground.**
2. no safer to use and ungrounded.
3. more dangerous to use and connected to ground.
4. safer to use and ungrounded.

Topic 18: Direct-Current Circuits

TOPIC DISCUSSION

Sources of emf



$$\Delta V = \mathcal{E} - IR$$

$$\Delta V = IR \quad \rightarrow \quad \mathcal{E} = IR + Ir \quad \rightarrow \quad I = \frac{\mathcal{E}}{R + r}$$

$$I\mathcal{E} = I^2R + I^2r$$

Think – Pair – Share 1

True or False: While discharging, the terminal voltage of a battery can never be greater than the emf of the battery.

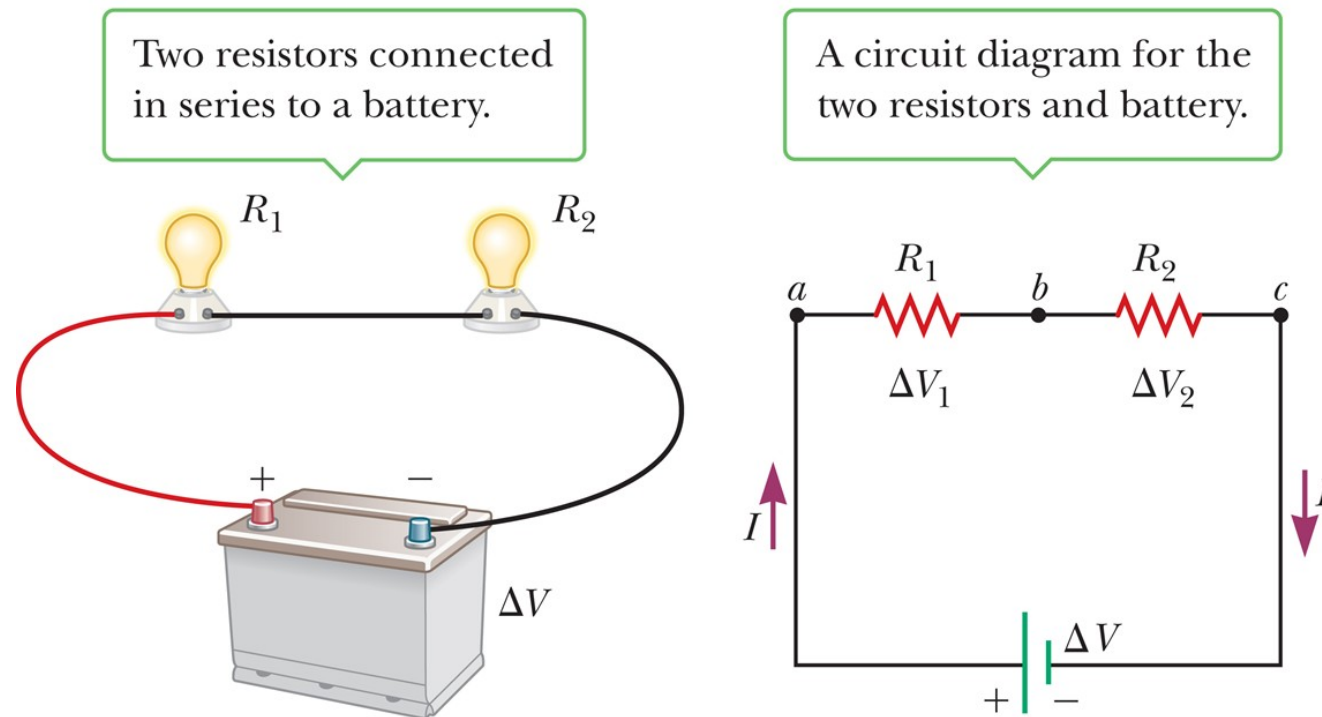
1. False
2. True

Think – Pair – Share 2

Why does a battery get warm while in use?

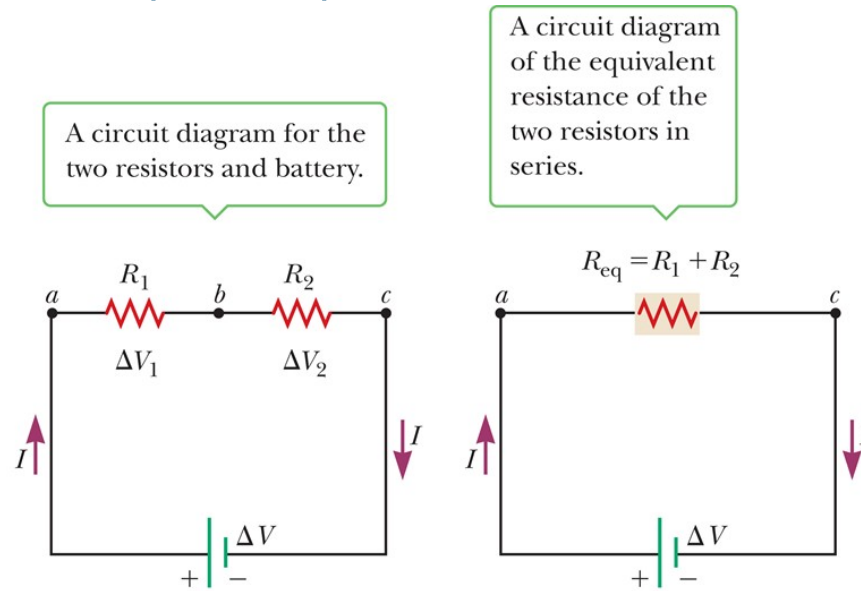
1. Because of internal resistance, power is delivered to the battery material, raising its temperature.
2. The electrons in the wire build up inside the battery, thus heating it.
3. Batteries actually cool down when in use.

Resistors in Series (1 of 2)



$$\Delta V = IR_1 + IR_2 = I(R_1 + R_2)$$

Resistors in Series (2 of 2)



$$\Delta V = IR_{\text{eq}}$$

$$IR_{\text{eq}} = I(R_1 + R_2) \quad \text{or}$$

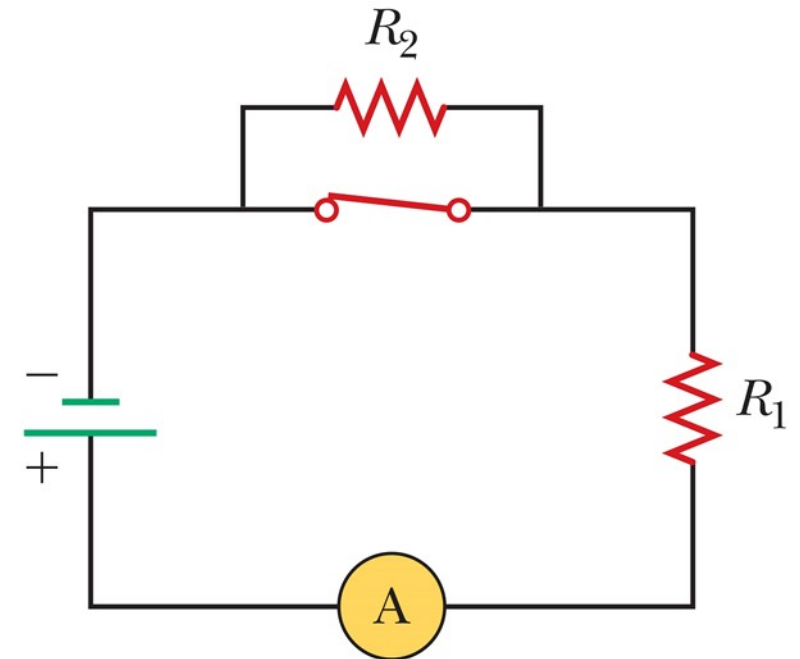
$$R_{\text{eq}} = R_1 + R_2 \quad (\text{series combination})$$

$$R_{\text{eq}} = R_1 + R_2 + R_3 + \dots$$

Think – Pair – Share 3

In the figure below, the current is measured with the ammeter at the bottom of the circuit. When the switch is opened, the reading on the ammeter

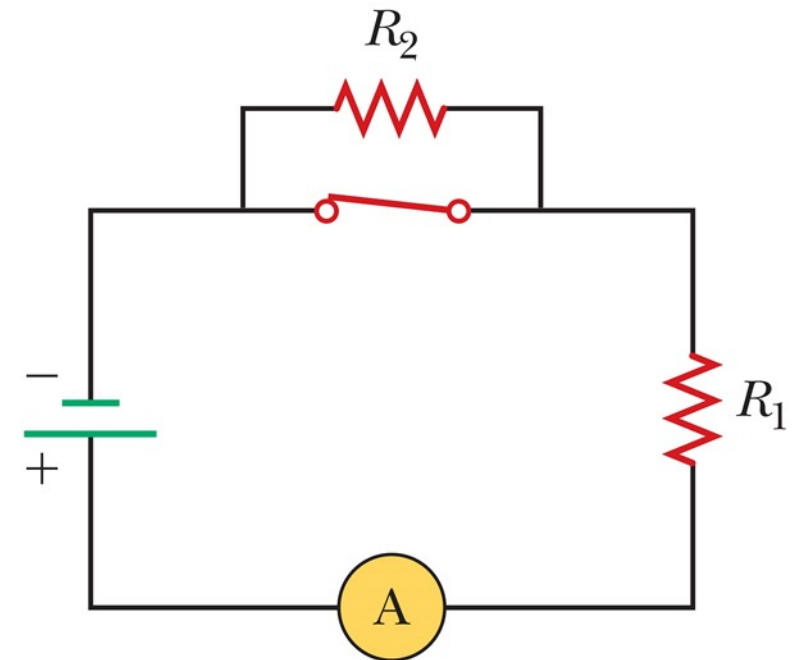
1. increases.
2. decreases.
3. does not change.



Think – Pair – Share 4

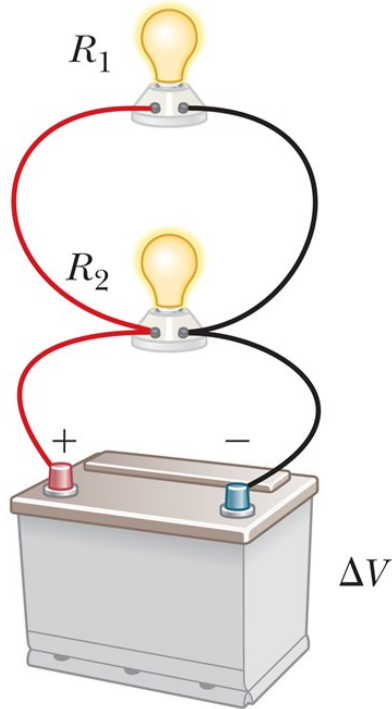
The circuit in the figure below consists of two resistors, a switch, an ammeter, and a battery. When the switch is closed, power P_c is delivered to resistor R_1 . When the switch is opened, which of the following statements is true about the power P_o delivered to R_1 ?

1. $P_o < P_c$
2. $P_o = P_c$
3. $P_o > P_c$



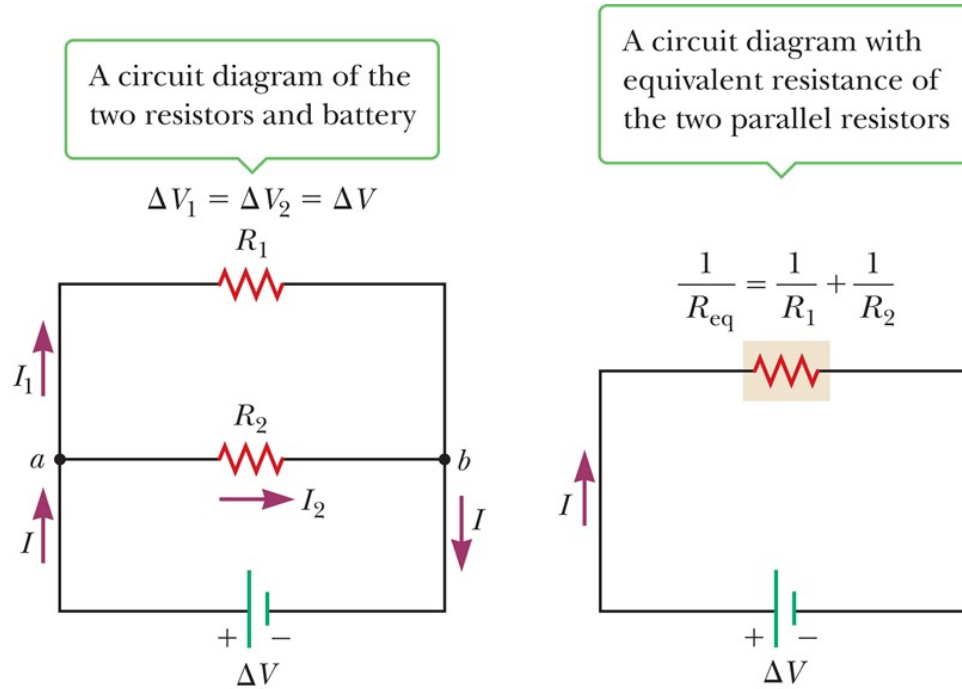
Resistors in Parallel (1 of 3)

Two resistors connected
in parallel to a battery



Resistors in parallel: the potential differences across the resistors are the same

Resistors in Parallel (2 of 3)



$$I = I_1 + I_2$$

$$I_1 = \frac{\Delta V}{R_1} \quad \text{and} \quad I_2 = \frac{\Delta V}{R_2}$$

$$I = \frac{\Delta V}{R_{eq}}$$

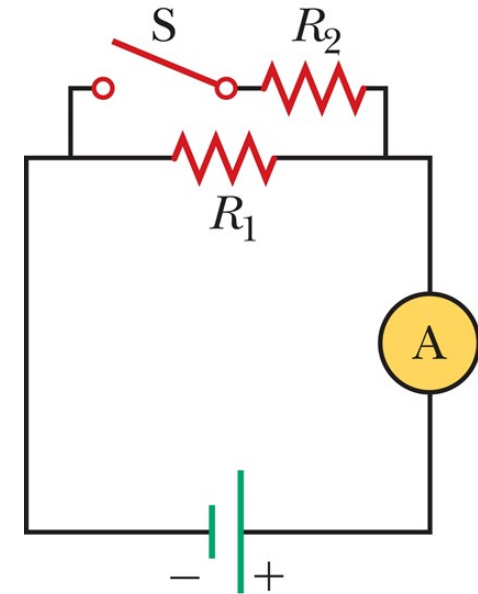
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots \quad (\text{parallel combination})$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

Think – Pair – Share 5

In the figure below, the current is measured with the ammeter on the right side of the circuit diagram. When the switch is closed, the reading on the ammeter

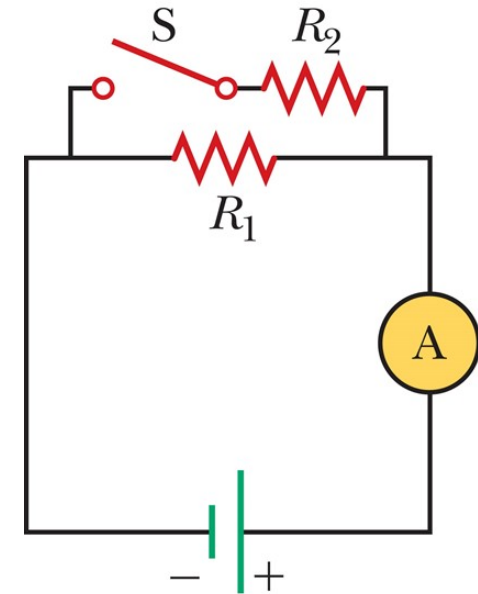
1. increases.
2. decreases.
3. remains the same.



Think – Pair – Share 6

When the switch is open in the figure below, power P_o is delivered to the resistor R_1 . When the switch is closed, which of the following is true about the power P_c delivered to R_1 ? (Neglect the internal resistance of the battery.)

1. $P_c < P_o$
2. $P_c = P_o$
3. $P_c > P_o$



Think – Pair – Share 7

Suppose you have three identical lightbulbs, some wire, and a battery. You connect one lightbulb to the battery and take note of its brightness. You add a second lightbulb, connecting it in parallel with the previous lightbulbs, and again take note of the brightness. Repeat the process with the third lightbulb, connecting it in parallel with the other two. As the lightbulbs are added, the brightness of the lightbulbs (neglect the battery's internal resistance)

1. increases.
2. decreases.
3. remains the same.
4. Need more information.

Think – Pair – Share 8

For the same scenario in the previous slide, as the lightbulbs are added, the individual currents in the lightbulbs

1. increases.
2. decreases.
3. remains the same.
4. Need more information.

Think – Pair – Share 9

Again, same scenario, as the lightbulbs are added, the power delivered by the battery

1. increases.
2. decreases.
3. remains the same.
4. Need more information.

Think – Pair – Share 10

Again, same scenario, as the lightbulbs are added, the lifetime of the battery

1. increases.
2. decreases.
3. remains the same.
4. Need more information.

Think – Pair – Share 11

Suppose you have three identical lightbulbs, some wire, and a battery. You connect one lightbulb to the battery and take note of its brightness. You add a second lightbulb, connecting it in series with the previous lightbulbs, and again take note of the brightness. Repeat the process with the third lightbulb, connecting it in series with the other two. As the lightbulbs are added, the brightness of the lightbulbs (neglect the battery's internal resistance)

1. increases.
2. decreases.
3. remains the same.
4. Need more information.

Think – Pair – Share 12

For the same scenario in the previous slide, as the lightbulbs are added, the individual currents in the lightbulbs

1. increases.
2. decreases.
3. remains the same.
4. Need more information.

Think – Pair – Share 13

Again, same scenario, as the lightbulbs are added, the power delivered by the battery

1. increases.
2. decreases.
3. remains the same.
4. Need more information.

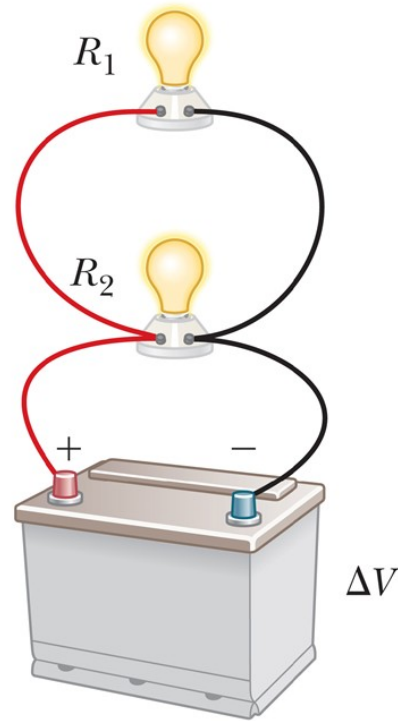
Think – Pair – Share 14

Again, same scenario, as the lightbulbs are added, the lifetime of the battery

1. increases.
2. decreases.
3. remains the same.
4. Need more information.

Resistors in Parallel (3 of 3)

Two resistors connected
in parallel to a battery



Problem-Solving Strategy: Simplifying Circuits with Resistors

1. Combine all resistors in series

- $R_{\text{eq}} = R_1 + R_2 + R_3 + \dots$
- The current in each resistor is the same.

2. Combine all resistors in parallel

- $\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$
- The potential difference across each resistor is the same.

3. Repeat

4. Use Ohm's law: $\Delta V = IR$

Kirchhoff's Rules and Complex DC Circuits (1 of 2)

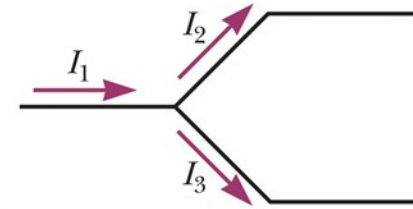
Kirchhoff's rules:

1. Junction rule

$$I_1 = I_2 + I_3$$

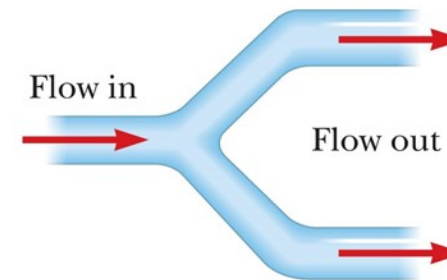
2. Loop rule

The current I_1 entering the junction must equal the sum of the currents I_2 and I_3 leaving the junction.



a

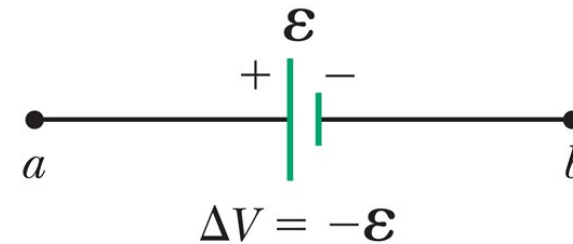
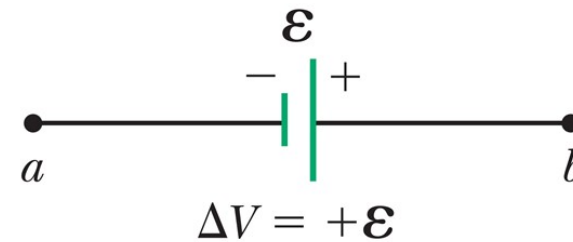
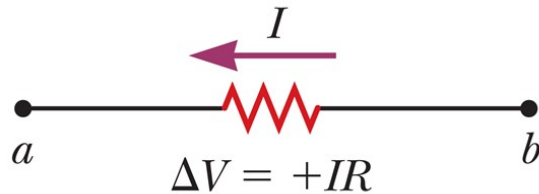
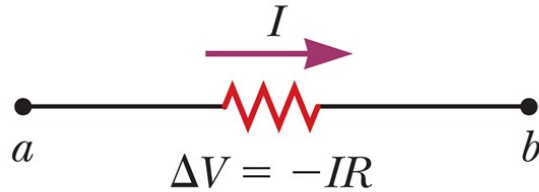
The net volume flow rate in must equal the net volume flow rate out.



b

Kirchhoff's Rules and Complex DC Circuits (2 of 2)

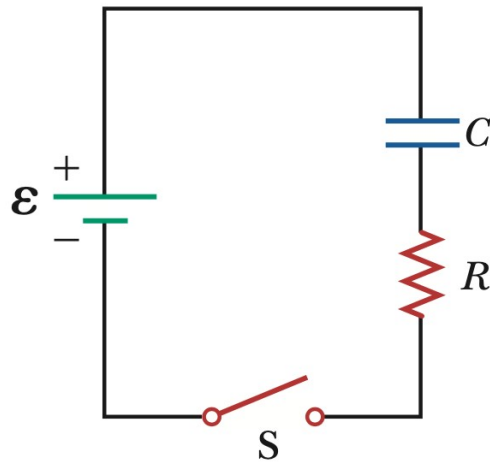
In each diagram, $\Delta V = V_b - V_a$
and the circuit element is
traversed from a to b , left to right.



Problem-Solving Strategy: Applying Kirchhoff's Rules to a Circuit

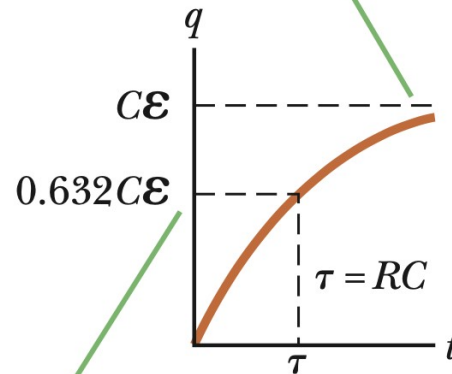
1. Assign labels and symbols
2. Assign *directions* to the currents
3. Apply the junction rule
4. Apply Kirchhoff's loop rule
5. Solve the equations
6. Check your answers

RC Circuits (1 of 3)



a

The charge approaches its maximum value $C\mathcal{E}$ as t approaches infinity.



After one time constant τ , the charge is 63.2% of the maximum value $C\mathcal{E}$.

b

$$Q = C\mathcal{E}$$

$$q = Q \left(1 - e^{-t/RC} \right)$$

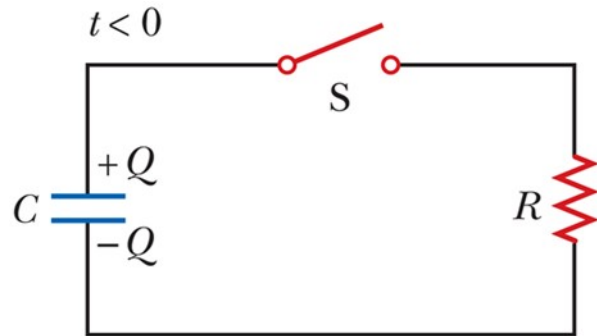
$$\Delta V = \frac{q}{C}$$

RC Circuits (2 of 3)

$$q = Q \left(1 - e^{-t/RC} \right)$$

$$q = Q \left(1 - e^{-RC/RC} \right) = Q \left(1 - e^{-1} \right) = 0.632 Q$$

RC Circuits (3 of 3)

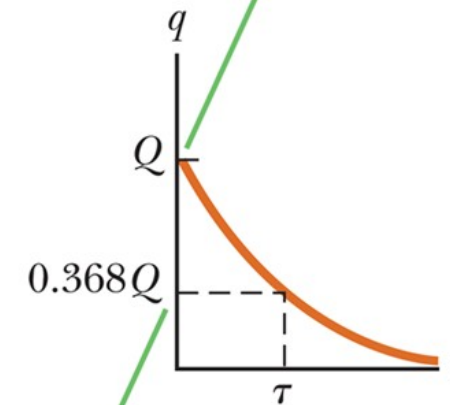


$$q = Q e^{-t/RC}$$

$$\Delta V = \frac{q}{C} \rightarrow \Delta V = \mathcal{E} e^{-t/RC}$$

$$\text{where } \mathcal{E} = \frac{Q}{C}$$

The charge has its maximum value Q at $t = 0$ and decays to zero exponentially as t approaches infinity.

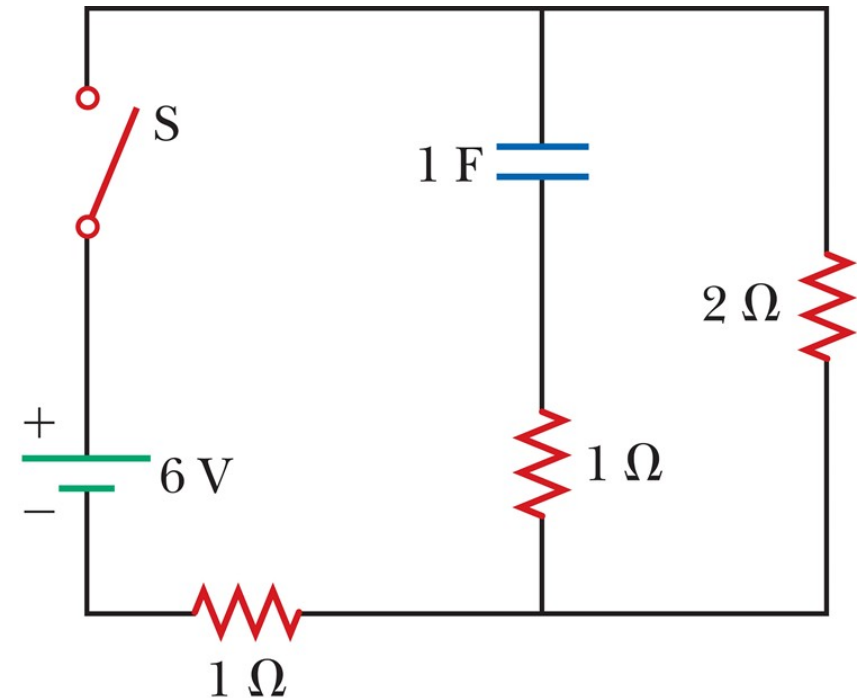


The charge drops to 36.8% of its initial value when one time constant has elapsed.

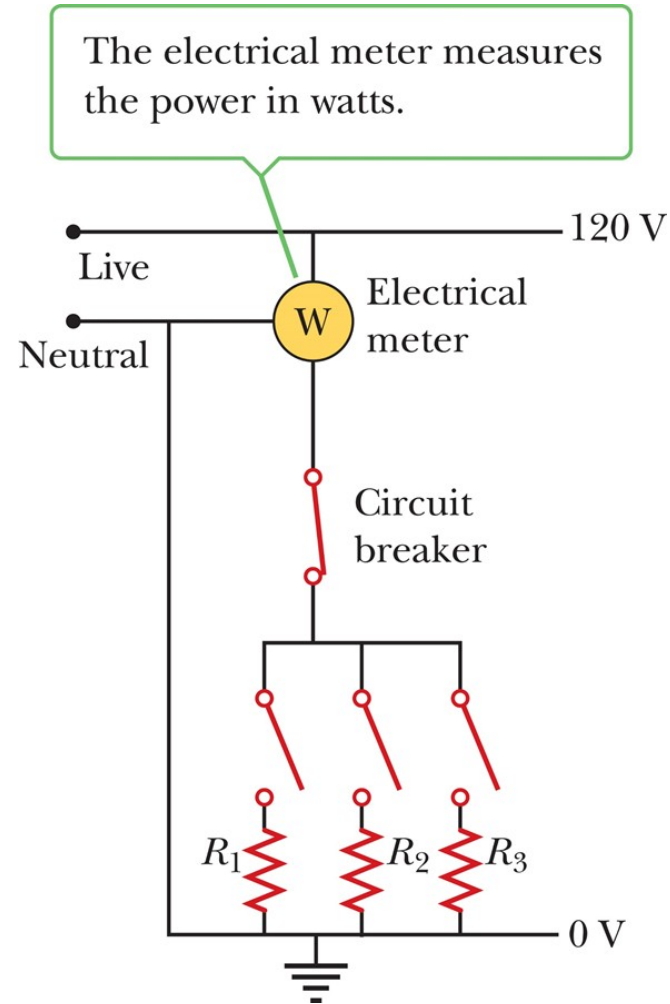
Think – Pair – Share 15

The switch is closed in the figure below. After a long time compared with the time constant of the circuit, what will the current be in the 2- Ω resistor?

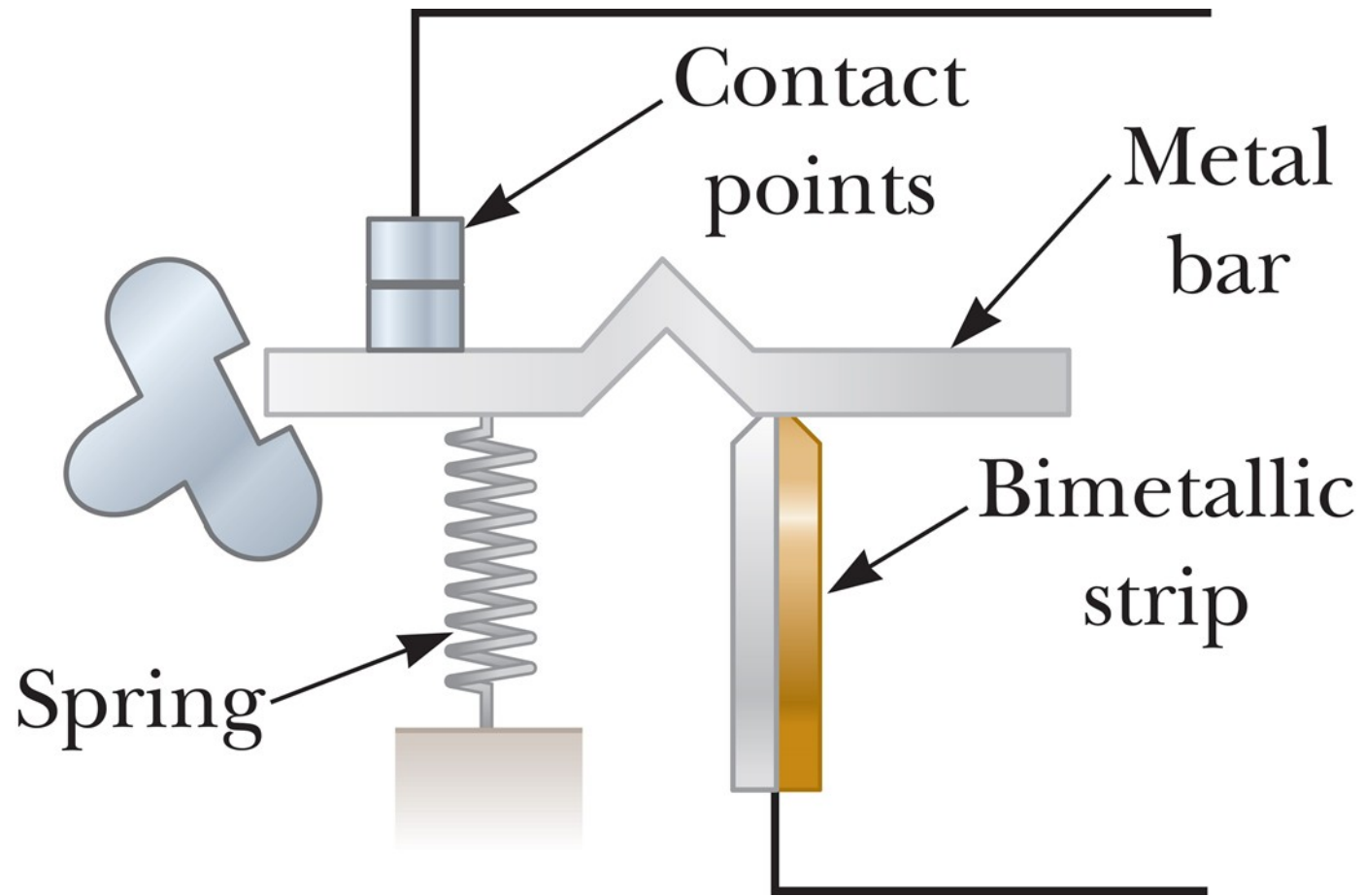
1. 4 A
2. 3 A
3. 2 A
4. 1 A



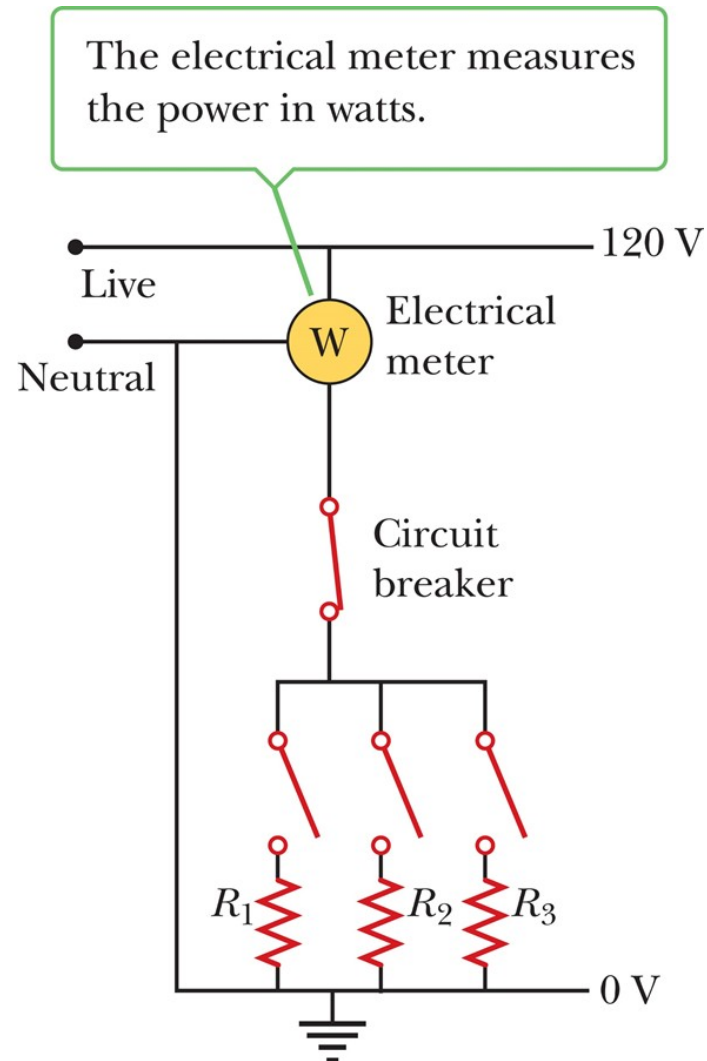
Household Circuits (1 of 4)



Household Circuits (2 of 4)



Household Circuits (3 of 4)



$$P = I \Delta V$$

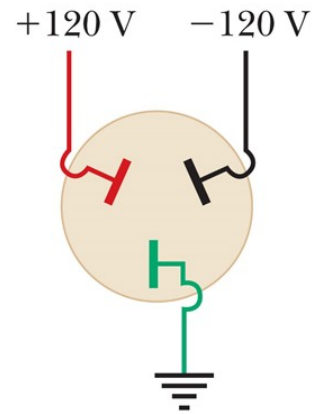
toaster: $I = \frac{P}{\Delta V} = \frac{1000 \text{ W}}{120 \text{ V}} = 8.33 \text{ A}$

microwave: $I = \frac{P}{\Delta V} = \frac{800 \text{ W}}{120 \text{ V}} = 6.67 \text{ A}$

heater: $I = \frac{P}{\Delta V} = \frac{1300 \text{ W}}{120 \text{ V}} = 10.8 \text{ A}$

$$I_{\text{total}} = 25.8 \text{ A}$$

Household Circuits (4 of 4)



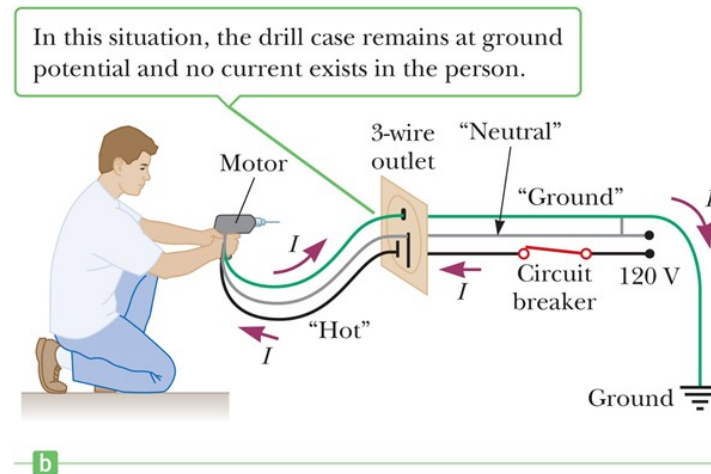
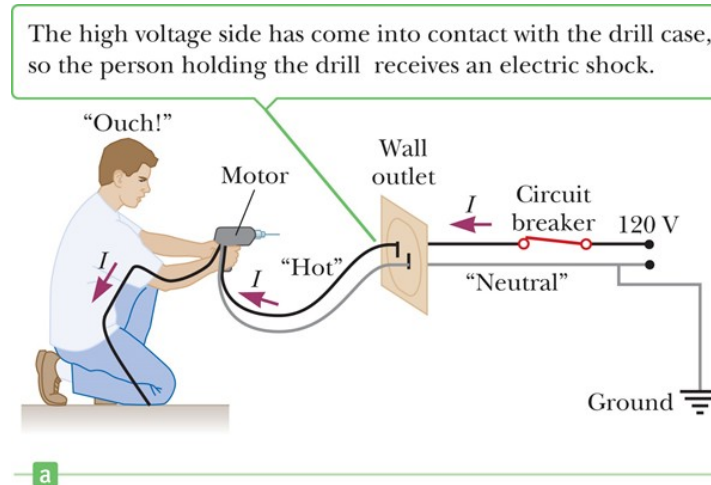
a

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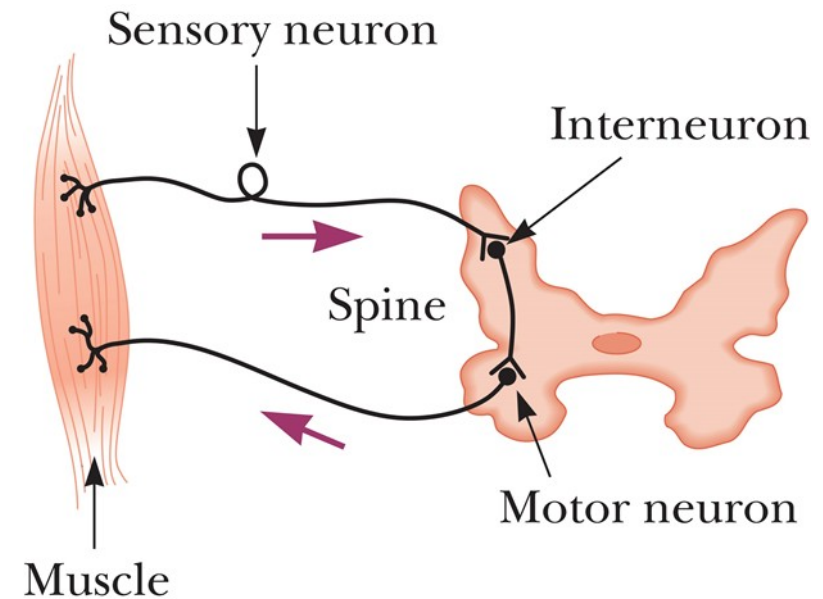
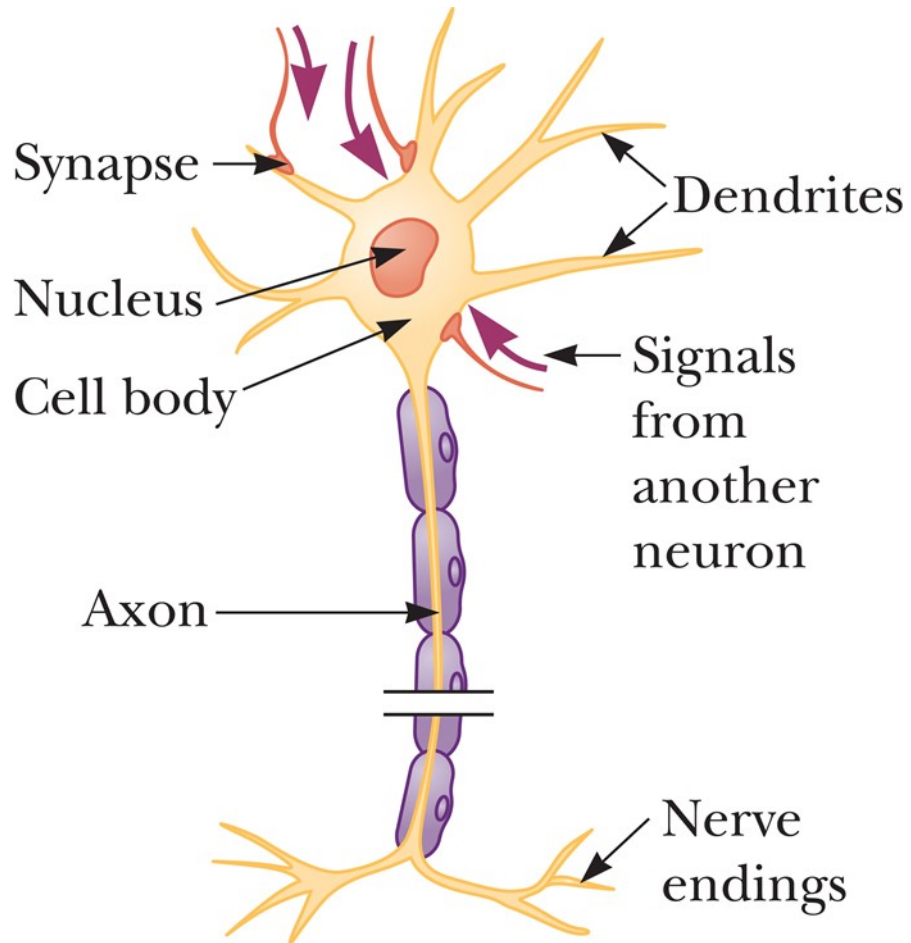


b

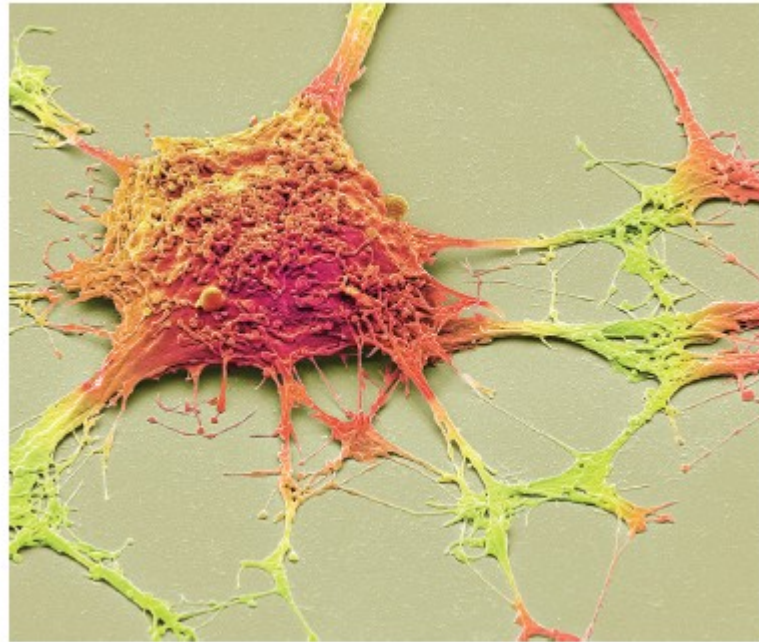
Electrical Safety



Conduction of Electrical Signals by Neurons (1 of 4)

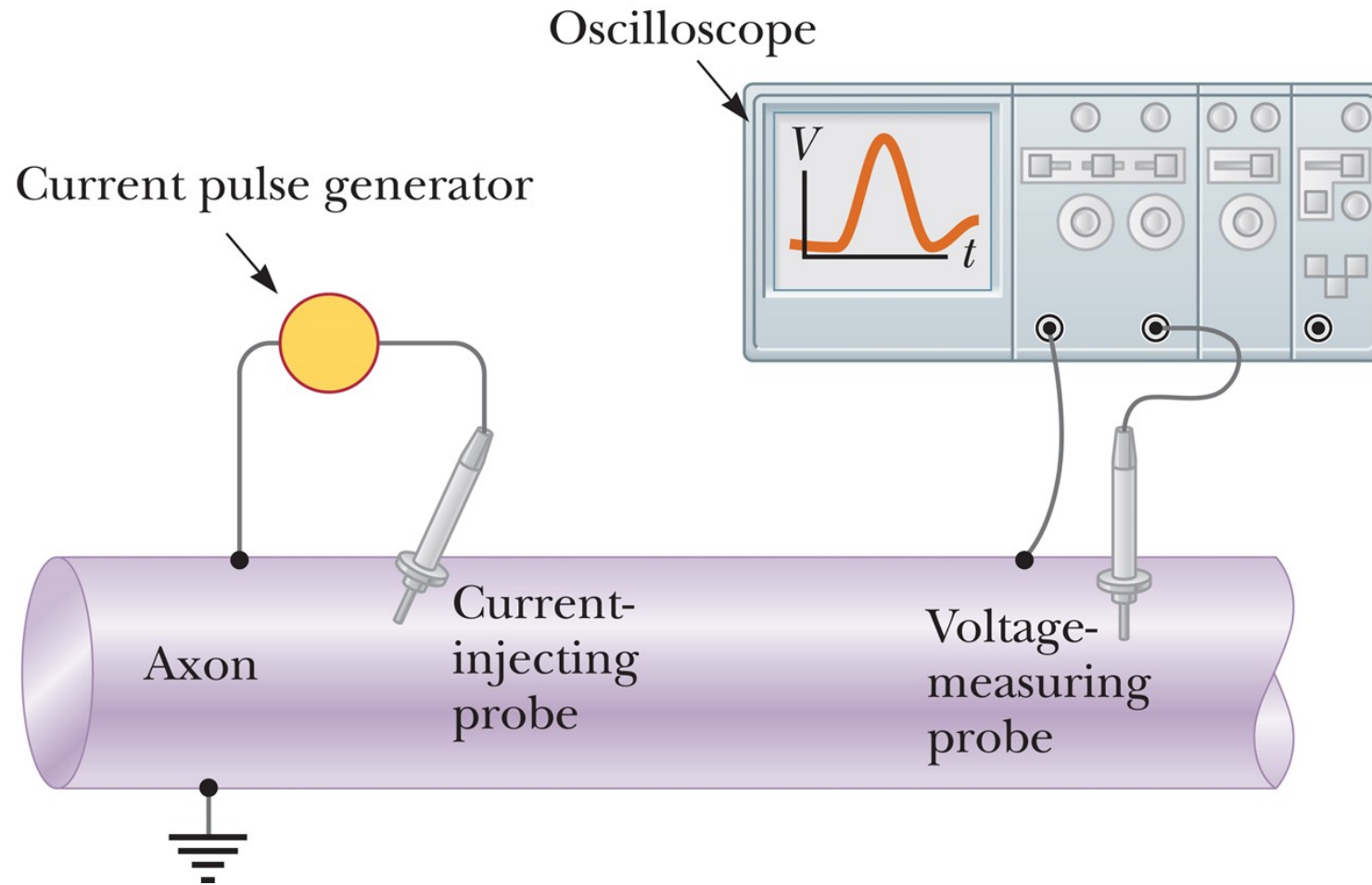


Conduction of Electrical Signals by Neurons (2 of 4)

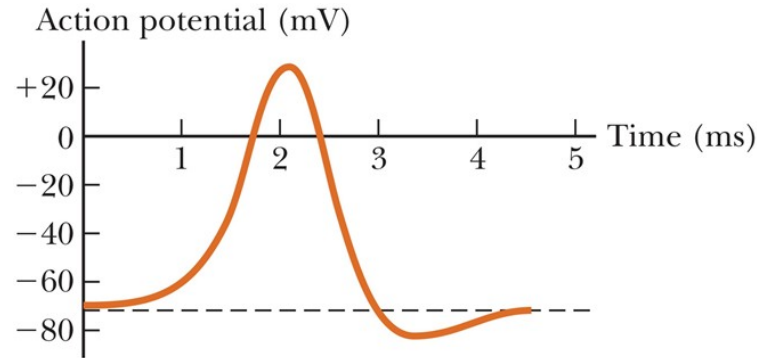


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Conduction of Electrical Signals by Neurons (3 of 4)

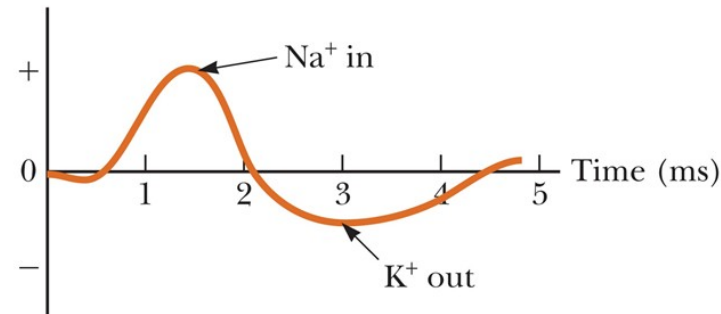


Conduction of Electrical Signals by Neurons (4 of 4)



a

Current in
membrane wall
(arbitrary units)



b